

LEARNING OBJECTIVES.

- *Build a Bayesian regression model and simulate its posterior.*
- *Conduct inference using Bayesian regression models.*
- *Evaluate Bayesian regression models.*

Categorical Predictors Let Z be a categorical variable with levels $1, \dots, k$. Then we can build a regression model with dummy/indicator variables $X_1, \dots, X_{k-1}$ where

$$X_i = \begin{cases} 1 & \text{if } Z = i \\ 0 & \text{otherwise} \end{cases}$$

The Bayesian regression model with a single binary predictor X is the same as before:

$$\begin{array}{ll} \text{Data:} & Y_i | \beta_0, \beta_1, \sigma \stackrel{\text{ind.}}{\sim} N(\mu_i, \sigma^2) \text{ with } \mu_i = \beta_0 + \beta_1 X_i \\ \text{Priors:} & \beta_0 \sim N(m_0, s_0^2) \\ & \beta_1 \sim N(m_1, s_1^2) \\ & \sigma \sim \text{Exp}(l) \end{array}$$

However, the interpretation of the coefficients is different:

Example 1. The `Galton` data set in the `mosaicData` package contains a set of measurements on parents and their adult children collected by Francis Galton in the late 1800s. We are interested in using the `sex` of the children to predict their `height`, in inches. Determine reasonable priors for each of the parameters, then fit a Bayesian regression model with these prior beliefs and the data. Summarize your results.

Multiple Regression

We can extend the regression model to one with p predictors, which can be a combination of quantitative and categorical predictors, and can also include higher order and interaction terms.

$$\begin{array}{ll}\text{Data:} & Y_i | \beta_0, \beta_1, \dots, \beta_p, \sigma \stackrel{\text{ind.}}{\sim} N(\mu_i, \sigma^2) \text{ with } \mu_i = \beta_0 + \beta_1 X_i + \dots + \beta_p X_p \\ \text{Priors:} & \beta_0 \sim N(m_0, s_0^2) \\ & \beta_1 \sim N(m_1, s_1^2) \\ & \dots \\ & \beta_p \sim N(m_p, s_p^2) \\ & \sigma \sim \text{Exp}(l)\end{array}$$

Example 2. Consider using the variables `sex`, `father`, `mother` to predict the `height` variable in the Galton data set. Decide on priors and then fit the model. Interpret the posterior credible intervals for the partial slope coefficients. Also calculate a 90% posterior predictive interval for the height of a daughter whose mother is 66 inches tall and whose father is 68 inches tall.

Example 3. Now fit a model that includes an interaction between `father` and `mother`. Determine if the interaction between the father and mother's height is significant.

Example 4. Now we want to evaluate and compare the three models from the previous examples. Run posterior predictive checks (using `pp_check`) on each to see if they produce similar data to the original. Then evaluate predictive accuracy using both the raw data and 10-fold cross-validation. Which do you think is the best overall model?

Bias-variance trade-off

A model is said to have **high bias** if it tends to be “far” from the observed relationship in the data; and **high variance** if estimates of the model significantly differ depending upon what data is used. In model building, there are trade-offs between bias and variability:

- Overly simple models with few or no predictors tend to have high bias but low variability (high stability).
- Overly complicated models with lots of predictors tend to have low bias but high variability (low stability).

Example 5. Consider the table below which compares 3 models (not those from our example). Discuss the differences in these models in terms of the bias-variance trade-off.

Model	Raw MAE	Cross-validated MAE
Model 1	2.79	2.83
Model 2	1.53	1.86
Model 3	1.06	2.12